Single-cell transcriptome shift of human traumatic brain injury

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Traumatic Brain Injury (TBI)

- External force
- Inertia
- Neuroinflammation

(Blennow et al, 2016)

Question: What are the molecular mechanisms contributing to neuroinflammation?

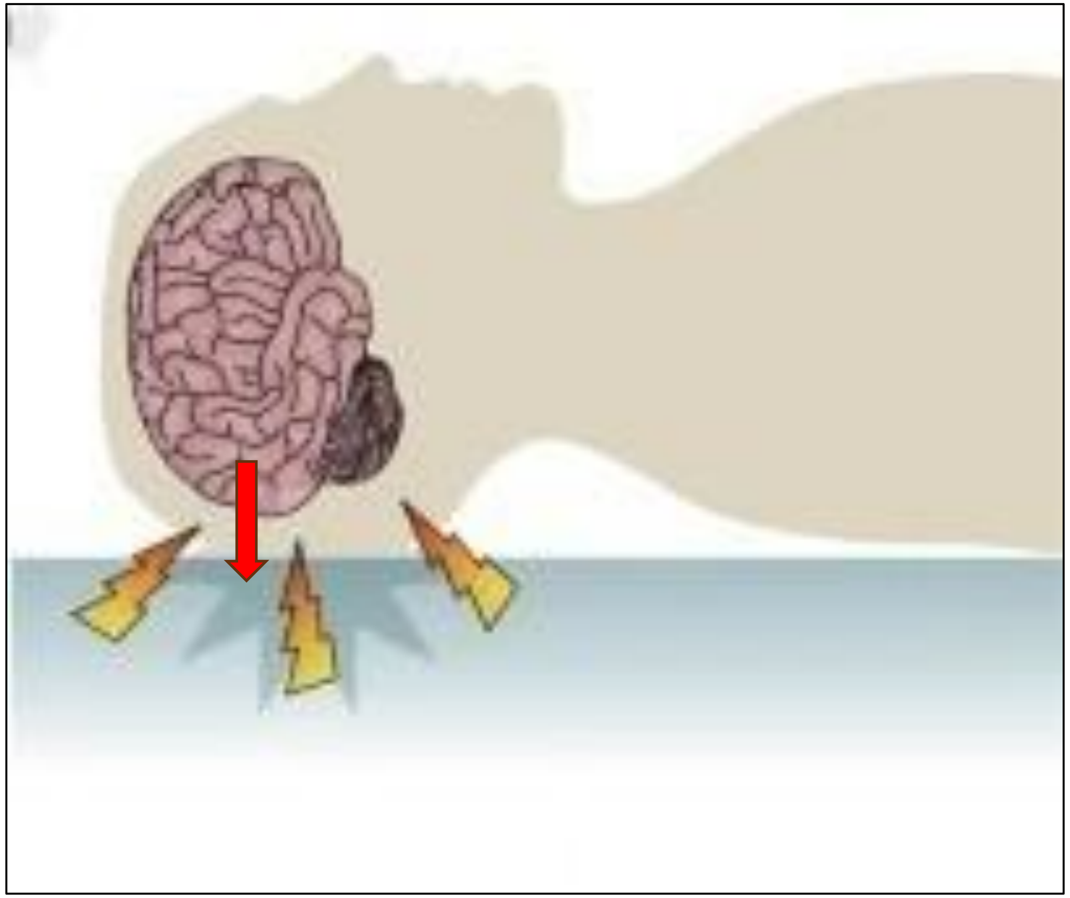

Figure 1: Physical mechanism of TBI
(Modified from Blennow et al, 2016; figure 1c)

Dataset

- GSE209552
- Single-nucleus RNA-seq
- **Control:** non-neuronal death postmortem brain
- **Treatment:** Surgically removed TBI sample

(Graze et al, 2023)

Sample	Region	Age	Sex	Injury	Time post injury (h)	GOSE
TBI 1	Temporal lobe	22	M	MVA	9	3
TBI 2	Frontal lobe	72	M	Fall	4	7
TBI 3	Frontal lobe	42	M	MVA	4	8
TBI 6	Temporal lobe	74	M	Fall	4	4
TBI 7	Temporal lobe	58	M	Fall	9	7
TBI 8	Temporal lobe	49	M	Fall	84	1
TBI 10	Temporal lobe	65	M	Fall	180	3
TBI 11	Parietal lobe	25	M	SPR	24	3
TBI 16	Temporal lobe	52	M	Fall	42	4
TBI 19	Frontal lobe	49	F	HOB	57	-
TBI 20	Frontal lobe	62	F	MVA	192	8
TBI 21	Frontal lobe	24	M	Fall	106	6
CTL 501	Frontal lobe	87	M	-	-	-
CTL 501	Temporal lobe	87	M	-	-	-
CTL 502	Temporal lobe	75	M	-	-	-
CTL 529	Frontal lobe	69	M	-	-	-
CTL 529	Temporal lobe	69	M	-	-	-

Figure 2: GSE209552 metadata overview; Yellow highlight tbi group; MVA = Motor vehicle accident, SPR = sports related, HOB = hit by object. GOSE = Glasgow Outcome Scale extended (Modified from Graze et al, 2023; supplemental table 1)

Quality Control

- Lower Mito % → Better quality
- Exclude GSM637381

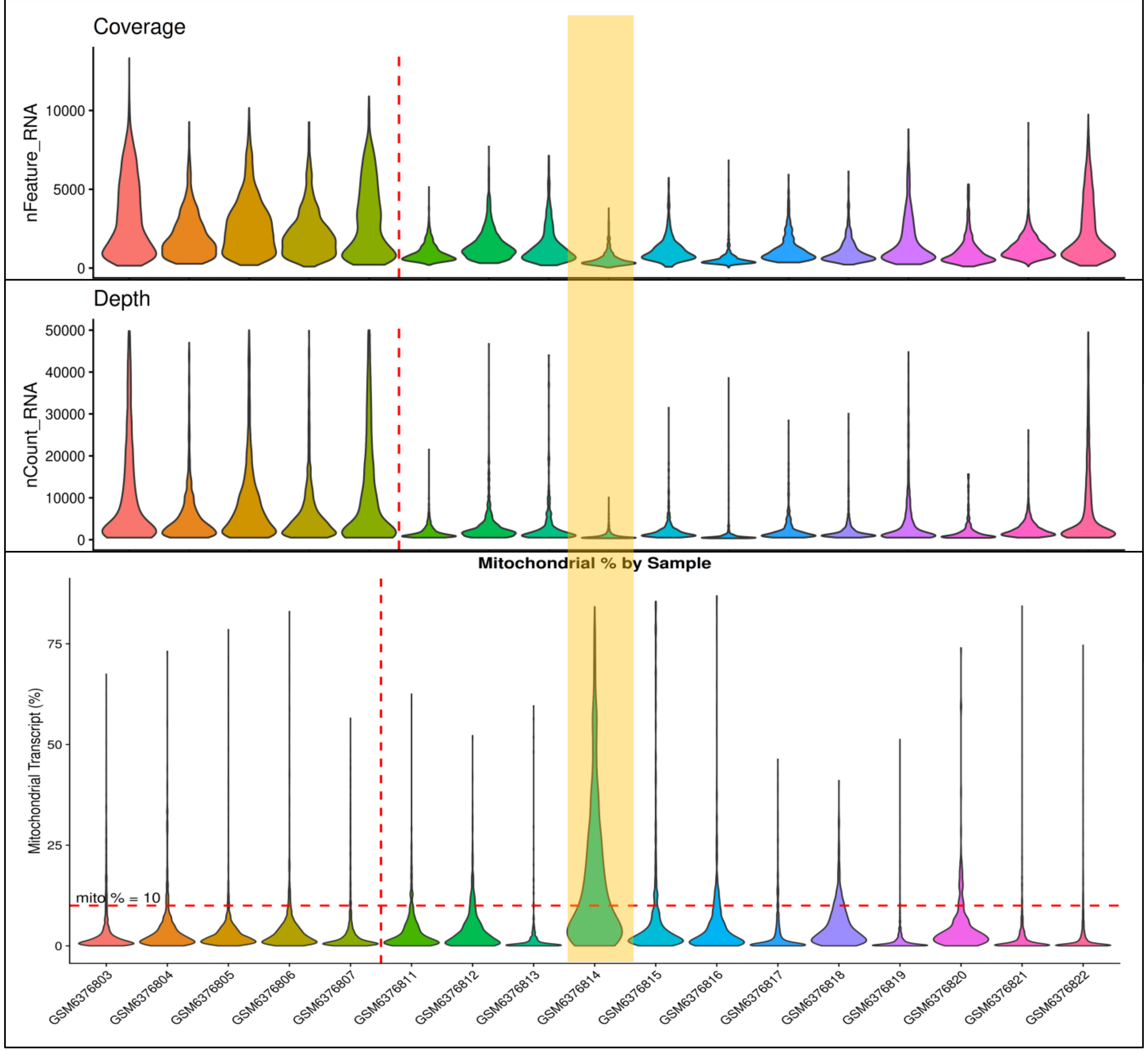

Figure 3: Sequence coverage, depth, and mitochondrial percentage per sample; The vertical red dash line separate control (left side of red line), and tbi (right side of red line)(Created using ggplot2, v4.0.1)

Normalization

- Log-CP10k
- Seurat v5.4.0
- NormalizeData()
  - Scale.factor = 10000
- Genes detected in less than 10 cells excluded

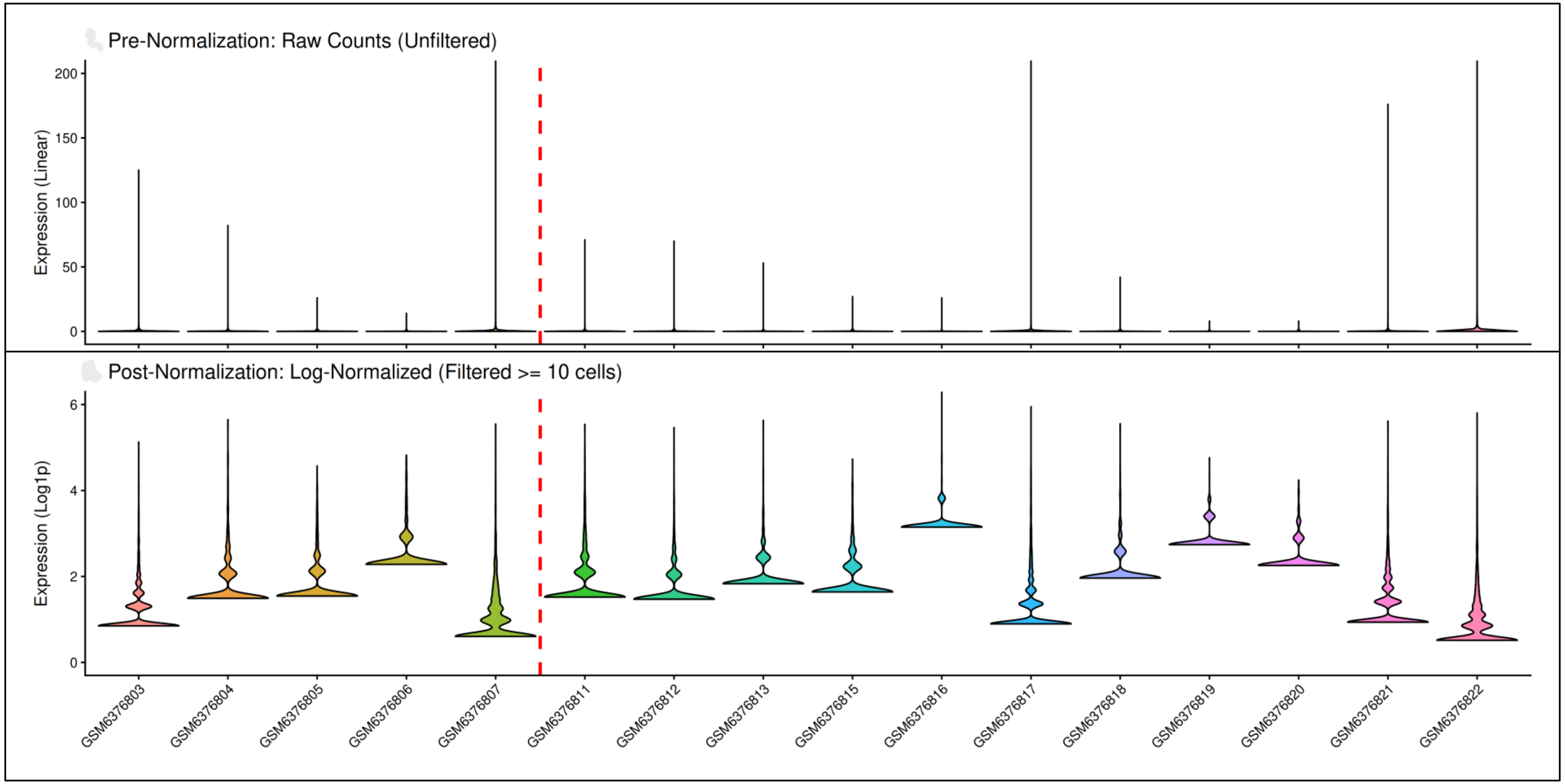

Figure 4: Pre and post normalized counts per cell (randomly sample 1 cell per GSM). The upper panel indicate the raw counts distribution; The bottom panel indicate the normalized counts after exclude all genes that expressed in less than 10 cells; X-axis indicate the GSMs that cell is sampled from; The vertical red dash line separate control and tbi)(Created using ggplot2, v4.0.1)

MDS Plot

- Principal component analysis (PCA)
- Seurat v5.4.0
- FindVariableFeatures()
    - Selection.method = “vst”
    - nfeature = 2000
- ScaleData()
- runPCA()
    - npcs = 20

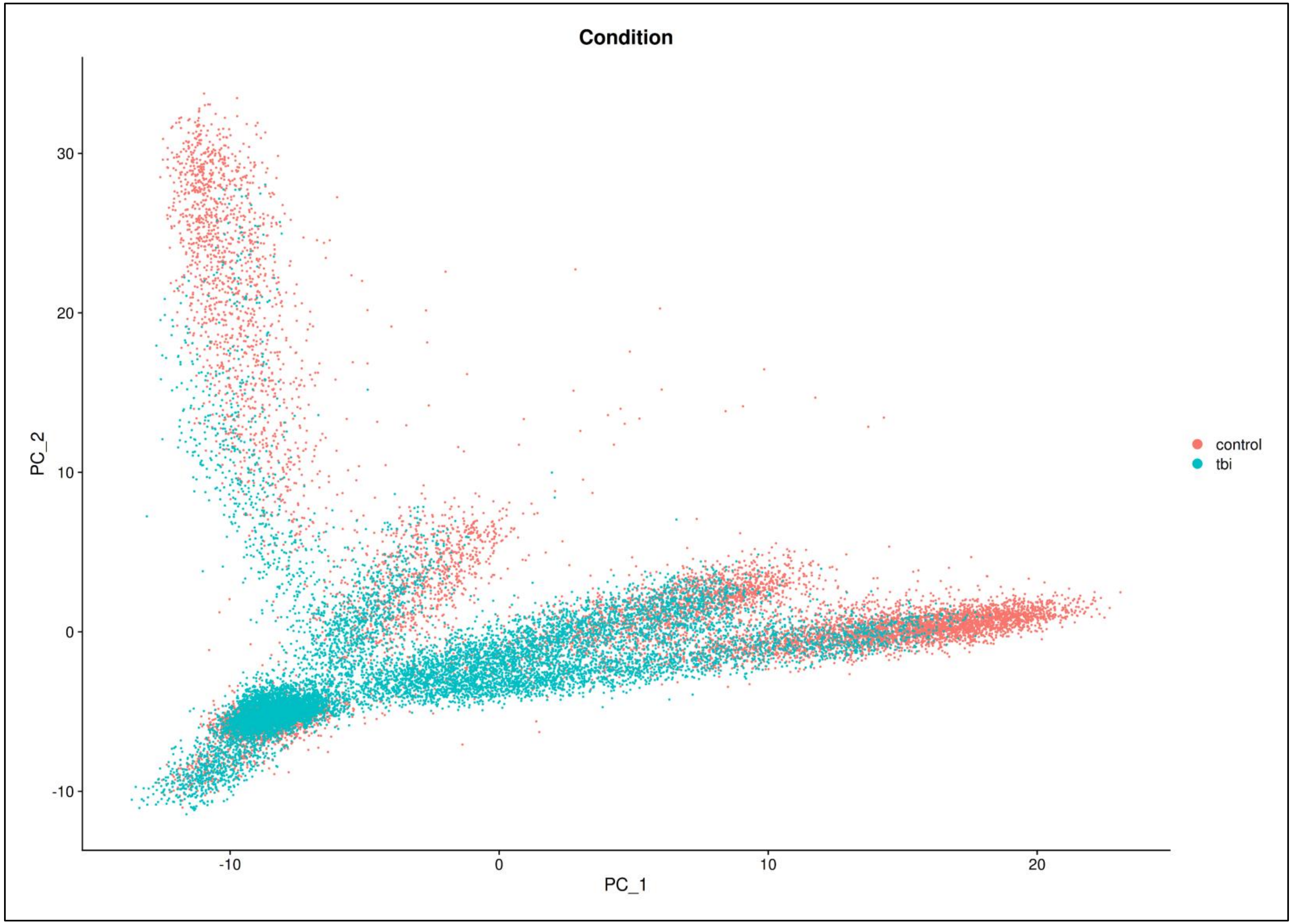

Figure 5: MDS plot grouped by condition. Y-axis is the second PC; X-axis is the first PC; Each dot indicate a single cell; Red denote control; Blue denote tbi (Created using Seurat, v5.4.0)

Differential Gene Expression

- Wilcoxon Rank-Sum test
- Bonferroni correction
- Seurat v5.4.0
- FindMarkers()

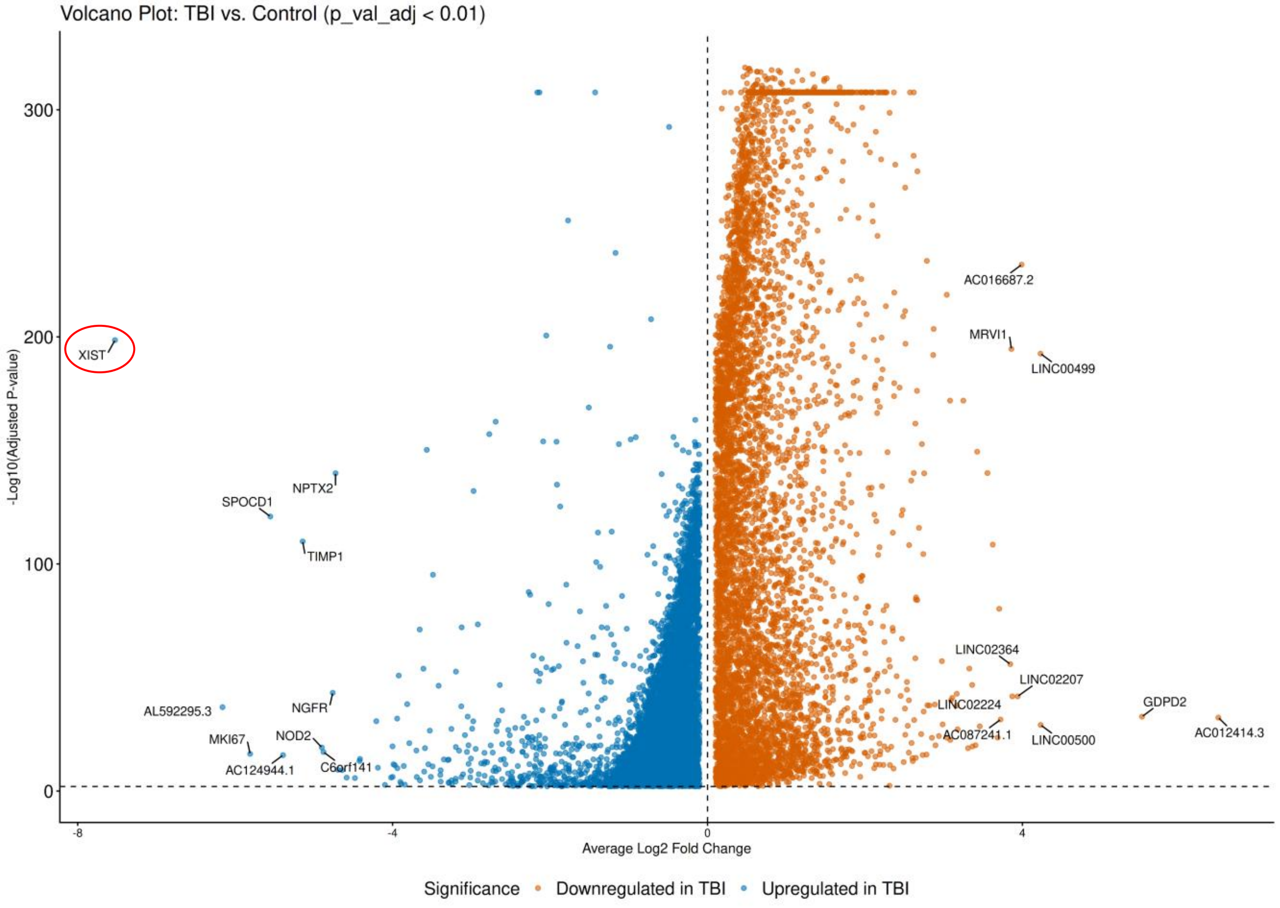

Figure 6: Volcano plot grouped by condition. Y-axis is negative log-transformed adjusted p value; X-axis is the average log-fold change centered at zero (Created using ggplot2, v4.0.1)

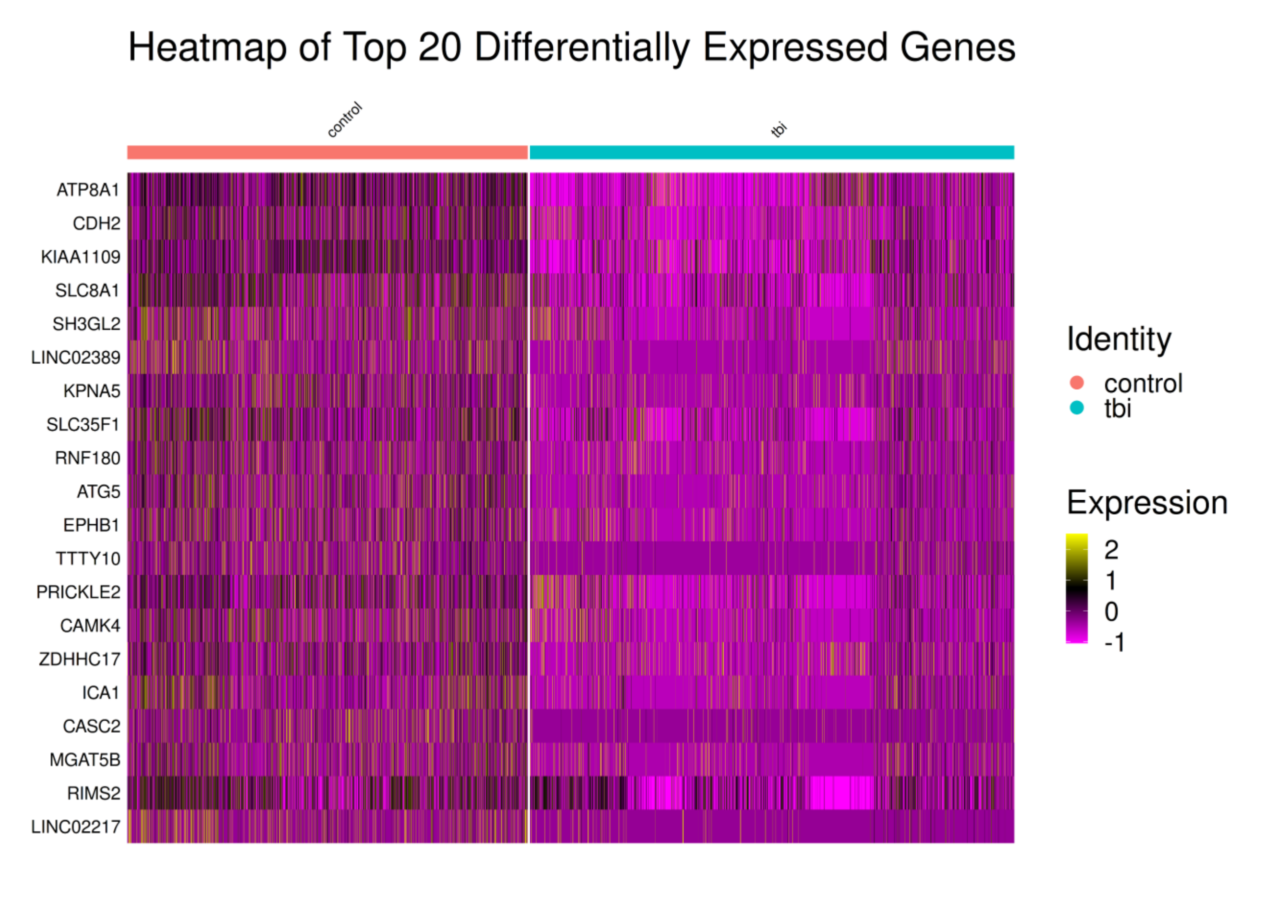

Figure 7: Heatmap of top 20 gene with lowest adjusted p value. Y-axis indicate the HOGO symbol of selected genes; Each tiny column along X-asix is a cell (Created using Seurat, v5.4.0)

Thresholded ORA

- Rank by $p\_FC = (1 - p\_val\_adj) \times (avg\_log2\_FC)$
- gprofiler2 v0.2.4
- gost()
  - correction_method = “fdr”
  - sources = “GO:BP” (e113_eg59_p19_6be52918)

Result (number of pathway identified):

- All: 496
- Up: 579
- Down: 112

term_id	term_name	p_value	intersection_size
GO:0044238	primary metabolic process	2.63e-49	2986
GO:0008152	metabolic process	5.05e-49	3297
GO:0043170	macromolecule metabolic process	5.16e-42	2714
GO:0009987	cellular process	7.07e-32	4650
GO:0071840	cellular component organization or biogenesis	1.95e-28	2008
GO:0009058	biosynthetic process	3.38e-28	2308
GO:0006139	nucleobase-containing compound metabolic process	1.07e-27	1779
GO:0006396	RNA processing	5.80e-27	393
GO:0016071	mRNA metabolic process	8.47e-26	309
GO:0046907	intracellular transport	9.30e-26	526

Figure 8: Top 10 significantly Up-regulated GO:BP pathway

term_id	term_name	p_value	intersection_size
GO:0120036	plasma membrane bounded cell projection organization	1.05e-16	388
GO:0030030	cell projection organization	2.03e-16	390
GO:0007399	nervous system development	8.12e-16	508
GO:0048731	system development	8.46e-12	723
GO:0099536	synaptic signaling	8.46e-12	188
GO:0099537	trans-synaptic signaling	1.48e-11	180
GO:0098916	anterograde trans-synaptic signaling	8.08e-11	176
GO:0007268	chemical synaptic transmission	8.08e-11	176
GO:0048699	generation of neurons	1.03e-10	311
GO:0048812	neuron projection morphogenesis	1.39e-10	161

Figure 9: Top 10 significantly Down-regulated GO:BP pathway

Non-thresholded GSEA

- GSEA Desktop v4.4.0
- Pathway data: Bader lab EM_Genesets (September_01_2025)
- Number of permutations: 2000
- Max size: 500 (default)
- Min size: 15 (default)

Result (number of pathway identified):

- Up: 4330
- Down: 1243

Table: Gene sets enriched in phenotype na [plain text format]

	GS follow link to MSigDB	GS DETAILS	SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val	RANK AT MAX	LEADING EDGE
1	HALLMARK_TNFA_SIGNALING_VIA_NFKB%MSIGDBHALLMARK%HALLMARK_TNFA_SIGNALING_VIA_NFKB	Details ...	119	-0.70	-2.82	0.000	0.000	0.000	1850	tags=37%, list=12%, signal=42%
2	PROTEIN SYNTHESIS: VALINE%PATHWHIZ%PW120528	Details ...	74	-0.75	-2.79	0.000	0.000	0.000	3133	tags=49%, list=20%, signal=61%
3	PROTEIN SYNTHESIS: LEUCINE%SMPDB%SMP0111873	Details ...	74	-0.75	-2.79	0.000	0.000	0.000	3133	tags=49%, list=20%, signal=61%
4	PROTEIN SYNTHESIS: CYSTEINE%PATHWHIZ%PW112918	Details ...	74	-0.75	-2.79	0.000	0.000	0.000	3133	tags=49%, list=20%, signal=61%
5	PROTEIN SYNTHESIS: TYROSINE%PATHWHIZ%PW120527	Details ...	74	-0.75	-2.79	0.000	0.000	0.000	3133	tags=49%, list=20%, signal=61%
6	PROTEIN SYNTHESIS: GLUTAMINE%SMPDB%SMP0111862	Details ...	73	-0.75	-2.78	0.000	0.000	0.000	3133	tags=48%, list=20%, signal=60%
7	PROTEIN SYNTHESIS: TRYPTOPHAN%PATHWHIZ%PW120526	Details ...	74	-0.75	-2.78	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
8	PROTEIN SYNTHESIS: HISTIDINE%PATHWHIZ%PW112929	Details ...	74	-0.75	-2.78	0.000	0.000	0.000	3139	tags=49%, list=20%, signal=61%
9	PROTEIN SYNTHESIS: SERINE%PATHWHIZ%PW120517	Details ...	73	-0.75	-2.77	0.000	0.000	0.000	3133	tags=48%, list=20%, signal=60%
10	PROTEIN SYNTHESIS: ALANINE%PATHWHIZ%PW101384	Details ...	74	-0.75	-2.77	0.000	0.000	0.000	3133	tags=49%, list=20%, signal=61%
11	PROTEIN SYNTHESIS: GLUTAMIC ACID%PATHWHIZ%PW112922	Details ...	74	-0.74	-2.77	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
12	PROTEIN SYNTHESIS: PROLINE%PATHWHIZ%PW113695	Details ...	74	-0.74	-2.77	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
13	PROTEIN SYNTHESIS: ASPARAGINE%SMPDB%SMP0111854	Details ...	74	-0.73	-2.76	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
14	EUKARYOTIC TRANSLATION ELONGATION%REACTOME%R-HSA-156842.4	Details ...	84	-0.72	-2.76	0.000	0.000	0.000	3176	tags=46%, list=21%, signal=58%
15	PROTEIN SYNTHESIS: GLYCINE%PATHWHIZ%PW112928	Details ...	74	-0.74	-2.76	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
16	CAP-DEPENDENT TRANSLATION INITIATION%REACTOME%R-HSA-72737.4	Details ...	107	-0.69	-2.75	0.000	0.000	0.000	3882	tags=52%, list=25%, signal=70%
17	PROTEIN SYNTHESIS: THREONINE%PATHWHIZ%PW120525	Details ...	74	-0.73	-2.75	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
18	PROTEIN SYNTHESIS: ASPARTIC ACID%SMPDB%SMP0111858	Details ...	74	-0.74	-2.74	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
19	PROTEIN SYNTHESIS: ARGININE%SMPDB%SMP0111853	Details ...	74	-0.73	-2.74	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%
20	PROTEIN SYNTHESIS: ISOLEUCINE%SMPDB%SMP0111872	Details ...	74	-0.74	-2.74	0.000	0.000	0.000	3133	tags=47%, list=20%, signal=59%

: TNF-α, NF-κB

: Protein synthesis

: Translation initiation & elongation

Figure 10: Top 20 significantly Up-regulated pathway

Table: Gene sets enriched in phenotype na ***[plain text format]***

	GS follow link to MSigDB	GS DETAILS	SIZE	ES	NES	NOM p-val	FDR q-val	FWER p-val	RANK AT MAX	LEADING EDGE
1	DETECTION OF STIMULUS INVOLVED IN SENSORY PERCEPTION%GOBP%GO:0050906	Details ...	46	0.63	2.13	0.000	0.009	0.021	2246	tags=46%, list=15%, signal=53%
2	DETECTION OF CHEMICAL STIMULUS INVOLVED IN SENSORY PERCEPTION%GOBP%GO:0050907	Details ...	20	0.75	2.09	0.000	0.013	0.061	2088	tags=60%, list=14%, signal=69%
3	EXPRESSION AND TRANSLOCATION OF OLFACTORY RECEPTORS%REACTOME%R-HSA-9752946.3	Details ...	22	0.73	2.06	0.000	0.014	0.097	2088	tags=55%, list=14%, signal=63%
4	SENSORY PERCEPTION OF SMELL%GOBP%GO:0007608	Details ...	24	0.71	2.05	0.000	0.012	0.109	3025	tags=63%, list=20%, signal=78%
5	CALCIUM-DEPENDENT CELL-CELL ADHESION VIA PLASMA MEMBRANE CELL ADHESION MOLECULES%GOBP%GO:0016339	Details ...	31	0.62	1.91	0.000	0.102	0.702	3605	tags=71%, list=23%, signal=93%
6	ASSEMBLY AND CELL SURFACE PRESENTATION OF NMDA RECEPTORS%REACTOME%R-HSA-9609736.5	Details ...	19	0.69	1.88	0.001	0.129	0.844	1745	tags=42%, list=11%, signal=47%
7	MOTILE CILIUM ASSEMBLY%GOBP%GO:0044458	Details ...	44	0.56	1.88	0.000	0.115	0.854	3089	tags=48%, list=20%, signal=60%
8	CARDIAC MUSCLE CELL ACTION POTENTIAL INVOLVED IN CONTRACTION%GOBP%GO:0086002	Details ...	31	0.60	1.86	0.002	0.132	0.917	2175	tags=48%, list=14%, signal=56%
9	DETECTION OF ABIOTIC STIMULUS%GOBP%GO:0009582	Details ...	58	0.53	1.86	0.000	0.121	0.924	2246	tags=33%, list=15%, signal=38%
10	GPCRS CLASS A RHODOPSIN LIKE%WIKIPATHWAYS_20250810%WP455%HOMO SAPIENS	Details ...	78	0.50	1.84	0.000	0.126	0.947	2185	tags=33%, list=14%, signal=39%
11	DETECTION OF EXTERNAL STIMULUS%GOBP%GO:0009581	Details ...	54	0.53	1.84	0.002	0.116	0.950	2246	tags=33%, list=15%, signal=39%
12	CATECHOL-CONTAINING COMPOUND METABOLIC PROCESS%GOBP%GO:0009712	Details ...	15	0.70	1.84	0.005	0.115	0.957	3131	tags=73%, list=20%, signal=92%
13	OLFACTORY SIGNALING PATHWAY%REACTOME%R-HSA-381753.8	Details ...	27	0.61	1.82	0.002	0.130	0.982	2088	tags=48%, list=14%, signal=56%
14	CATECHOLAMINE METABOLIC PROCESS%GOBP%GO:0006584	Details ...	15	0.70	1.81	0.003	0.130	0.988	3131	tags=73%, list=20%, signal=92%
15	SPERM MOTILITY%GOBP%GO:0097722	Details ...	86	0.48	1.80	0.000	0.142	0.997	4451	tags=55%, list=29%, signal=77%
16	SPERM FLAGELLUM ASSEMBLY%GOBP%GO:0120316	Details ...	31	0.59	1.79	0.004	0.144	0.998	2798	tags=52%, list=18%, signal=63%
17	HEART CONTRACTION%GOBP%GO:0060047	Details ...	49	0.53	1.79	0.002	0.140	0.998	2175	tags=43%, list=14%, signal=50%
18	CILIUM MOVEMENT%GOBP%GO:0003341	Details ...	123	0.45	1.79	0.000	0.133	0.999	3640	tags=46%, list=24%, signal=60%
19	FLAGELLATED SPERM MOTILITY%GOBP%GO:0030317	Details ...	86	0.48	1.79	0.001	0.129	0.999	4451	tags=55%, list=29%, signal=77%
20	PHASE 0 - RAPID DEPOLARISATION%REACTOME%R-HSA-5576892.4	Details ...	25	0.61	1.78	0.004	0.131	0.999	2025	tags=52%, list=13%, signal=60%

Figure 11: Top 20 significantly Down-regulated pathway :Sensory recep. :Olfactory recep. :Ion func. :Cilium, Sperm motility.

Network Visualization

- Cytoscape v3.10.4 (Shannon et al, 2003)
- EnrichmentMap v3.5.0 (Merico et al, 2010)
- AutoAnnotate v1.5.2 (Reimand et al, 2019)
- Node Cutoff (FDR q-value): 0.05
- Edge Cutoff (Similarity): 0.375

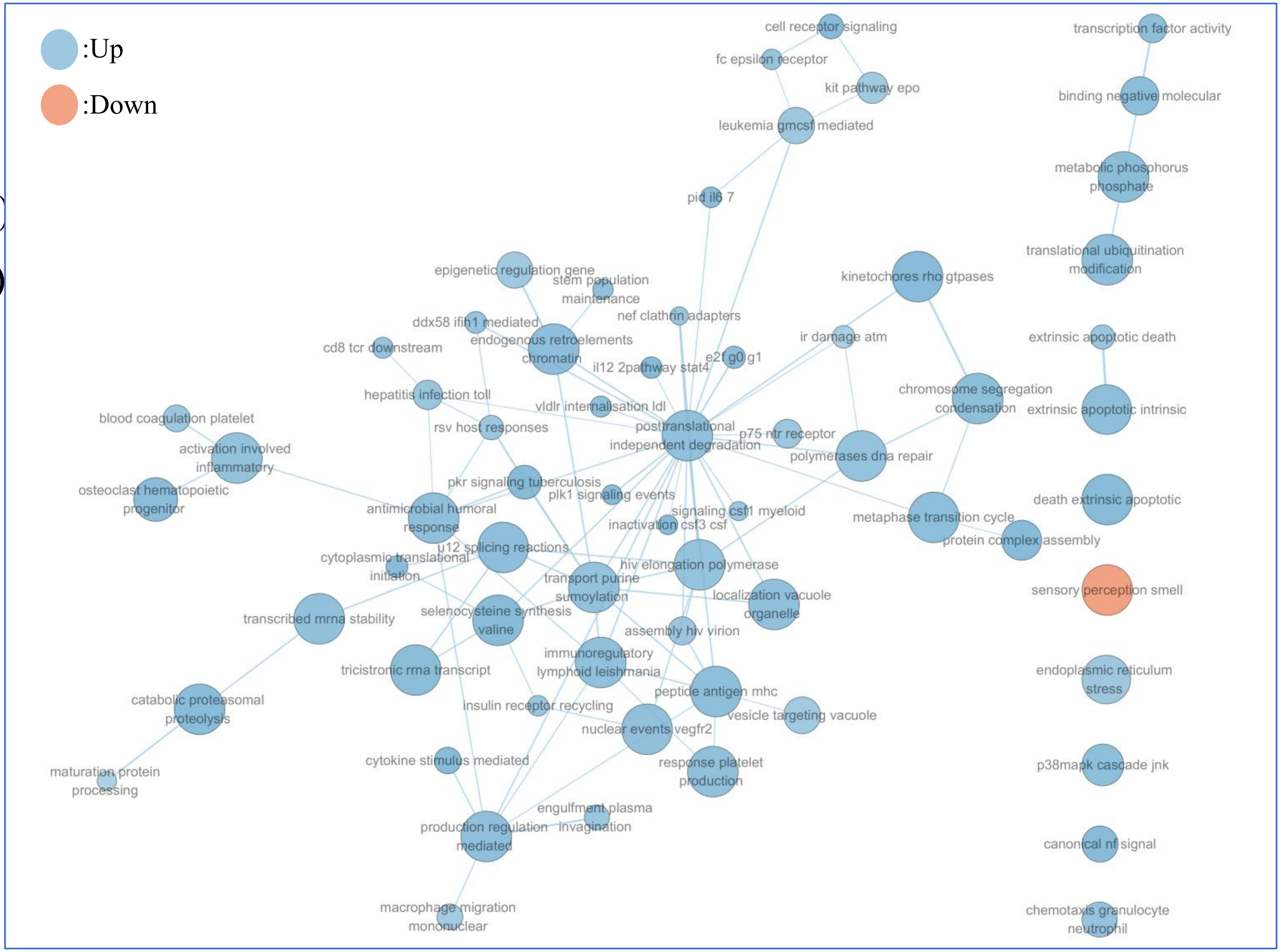

Figure 12: Collapsed Theme network (Created using Cytoscape, v3.10.4)

Discussion

- Known pathway/hallmark
  - TNF-α, NF-κB (Zhang et al, 2025, Longhi et al, 2013, Mettand et al, 2018)
  - Olfactory dysfunction (Liu et al, 2023, Luca et al, 2023, Xydakis et al, 2015)
  - Motile cilia impairment (Xing et al, 2022, Xiong et al, 2014, Gupta et al, 2024, Nelles & Hazrati, 2023)
- Noval pathway
  - None
- Study design
  - Weakly controlled variables (sex, age, time post injury)

Reference

- Pivotal

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    - “Create Elegant Data Visualisations Using the Grammar of Graphics.” n.d. Accessed February 13, 2026. https://ggplot2.tidyverse.org/. Gprofiler2
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    - Merico, Daniele, Ruth Isserlin, Oliver Stueker, Andrew Emili, and Gary D. Bader. 2010. “Enrichment Map: A Network-Based Method for Gene-Set Enrichment Visualization and Interpretation.” *PLOS ONE* 5 (11): e13984. https://doi.org/10.1371/journal.pone.0013984.

- Exhaustive reference list
    - https://github.com/bcb420-2026/Jiaqi_Ma/blob/main/assignment_2/BCB420.bib